

Om Mishra

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EXPERIENCE

FlyRank AI — Machine Learning Intern

July 2026 – September 2026 | Remote

- Built end-to-end machine learning workflows on real-world search and content data, covering data exploration, feature engineering, preprocessing, and model evaluation.
- Engineered a **90-day search visibility feature set** combining search performance, user engagement, content metadata, and content freshness signals across **82K+ content–client pairs**.
- Developed a **search visibility decline prediction and ranking system** using client-grouped validation, achieving **75% Precision@20 vs. 45% for the baseline**.
- Delivered a **Search Intelligence capstone** that converted model predictions into ranked content recommendations with interpretable reason codes and an actionable playbook, documented as a deployed research paper.

PROJECTS

Transformer Translation System



- Built and trained a complete Encoder-Decoder Transformer architecture from scratch for English-to-French neural machine translation on **2.3M sentence pairs** using PyTorch.
- Implemented SentencePiece tokenization, positional encoding, multi-head attention, masking, encoder-decoder cross-attention, beam search decoding, and autoregressive inference.
- Achieved a highly competitive **34.82 BLEU** score on **24k sentence validation set** and developed a Streamlit-based translation app with cross-attention heatmap visualizations.
- Improved training stability using Noam Learning Rate Scheduling, gradient clipping, custom validation pipelines, and numerical stability debugging during large-scale Transformer training.

PulmoScan AI: High-Sensitivity Pneumonia Screening Tool



- Engineered a medical diagnostic system to detect pneumonia from chest X-rays, explicitly prioritizing **Recall (Sensitivity)** to minimize false negatives in clinical screening.
- Fine-tuned and benchmarked DenseNet121 against ResNet50, achieving **99.7% Recall** versus **97.0%**, reducing missed pneumonia cases from 11 to 1.
- Implemented Explainable AI (Grad-CAM) with bicubic interpolation to visualize lung opacities and deployed a Streamlit application with caching, reducing inference latency by approximately **80%**.

Ramartha: Semantic & Narrative Analysis of Shri Ramcharitmanas



- Built an end-to-end NLP system for semantic analysis of a low-resource classical text using Word2Vec & FastText, with a Unicode-native preprocessing pipeline and verse-level embeddings for semantic retrieval & narrative-aware analysis, deployed through a Streamlit app for real-time exploration of semantic relationships & concept similarity.

Urban Mobility Optimization: Bike Demand Prediction



- Built an ML forecasting system for hourly bike demand, achieving **R2 = 0.935** & evaluated 7 regression models, selecting a tuned **LightGBM Regressor**, with cyclic feature engineering, log-transformation, & SHAP for model interpretability.

SUMMARY

Aspiring Machine Learning Engineer proficient in Python, Deep Learning, and Transformer architectures, focused on building end-to-end AI systems. Experienced in developing and deploying ML applications using PyTorch, Scikit-learn, and Hugging Face, with a strong emphasis on model evaluation, explainability, and real-world impact. Strong foundation in Data Structures, Algorithms, and software engineering best practices.

EDUCATION

Bachelor of Science in Information Technology

Smt. C.H.M. College

2024 – 2027 Ulhasnagar, Maharashtra

SKILLS

Machine Learning & Deep Learning

PyTorch, Scikit-learn, CNNs, Transfer Learning

Data Analysis & Visualization

NumPy, Pandas, Matplotlib, Seaborn, Plotly

Computer Vision & XAI

OpenCV, Grad-CAM, SHAP

Deployment & Tools

Streamlit, FastAPI, Git, GitHub, HuggingFace Hub, VS Code

CERTIFICATIONS

Oracle AI Foundations Associate

Certified in Artificial Intelligence fundamentals, Machine Learning concepts, and responsible AI by Oracle.

NASA Space Apps Challenge – Galactic Problem Solver

Recognized for contributing innovative solutions during NASA's global hackathon challenge.